4.6 The Fundamental Theorem of Algebra

Name _____ Period ____ Date _____

Algebra II 4.6 (30-minute plan) · Student sheet

I can…

- use the Fundamental Theorem of Algebra to find the number of solutions of a polynomial equation.
- find all the zeros of a polynomial function and write polynomials from their zeros.
- use Descartes's Rule of Signs.

Part 1 · Notes: fill in the blanks

A polynomial equation of degree n has exactly _____ solutions.

Imaginary zeros come in _____ pairs: a + bi and _____.

Descartes's Rule: count the sign changes in _____ and in _____.

Part 2 · Examples (work them with the class)

Example 1

How many solutions does $x^3 + 64x = 0$ have? Find them.

Example 2

Write a polynomial function f of least degree with leading coefficient 1 that has the zeros 3 and 2i.

Part 3 · Partner practice

Partner practice

Write a polynomial function of least degree with leading coefficient 1 that has the zeros 1 (repeated) and 4i.